

Transformation Evidence for Congruence and Similarity

Congruence from lengths, similarity from ratios, slope evidence, and naming the transformation

Directions

- Distance between two points: $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$. Slope: $\frac{y_2 - y_1}{x_2 - x_1}$.
- Congruent** figures have equal matching lengths. **Similar** figures have matching lengths in one constant ratio.
- The problems switch between the four kinds of evidence on purpose — decide which one the question needs before you calculate. All lengths are in grid units.

1. $A(1, 1)$ and $B(4, 5)$ are translated 5 units right, giving $A'(6, 1)$ and $B'(9, 5)$.

$$AB = \underline{\hspace{2cm}} \quad A'B' = \underline{\hspace{2cm}}$$

2. $C(1, 2)$ and $D(7, 10)$ are reflected in the y -axis, giving $C'(-1, 2)$ and $D'(-7, 10)$.

$$CD = \underline{\hspace{2cm}} \quad C'D' = \underline{\hspace{2cm}}$$

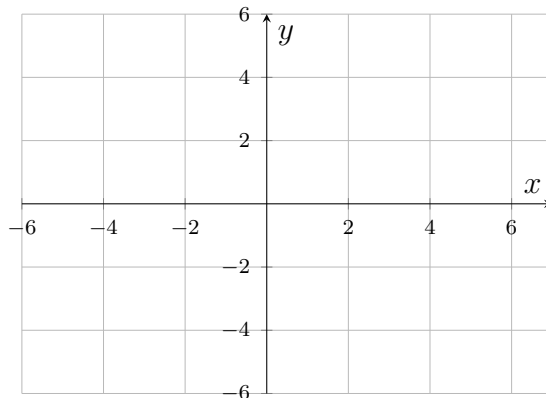
3. One triangle has sides 3, 4 and 5. A second triangle has sides 9, 12 and 15. Divide each larger side by the side it matches.

$$\frac{9}{3} = \underline{\hspace{2cm}} \quad \frac{12}{4} = \underline{\hspace{2cm}} \quad \frac{15}{5} = \underline{\hspace{2cm}}$$

4. $E(2, 5)$ and $F(6, 7)$ are rotated 180° about the origin, giving $E'(-2, -5)$ and $F'(-6, -7)$.

$$\text{Slope of } EF = \underline{\hspace{2cm}} \quad \text{Slope of } E'F' = \underline{\hspace{2cm}}$$

- 5.** Triangle P has vertices $(1, 2)$, $(5, 2)$ and $(1, 5)$. Triangle Q has vertices $(1, -2)$, $(5, -2)$ and $(1, -5)$. Both are drawn on the grid if you want to check your thinking.



Base of P = _____ Base of Q = _____

Name the single transformation that maps P onto Q , and explain how it shows the two triangles are congruent.

- 6.** A student says the triangle with vertices $(0, 0)$, $(6, 0)$, $(0, 4)$ is congruent to the triangle with vertices $(0, 0)$, $(3, 0)$, $(0, 2)$.

First base = _____ Second base = _____

Is the student right? Say what the two lengths show, and give the correct description of the pair.

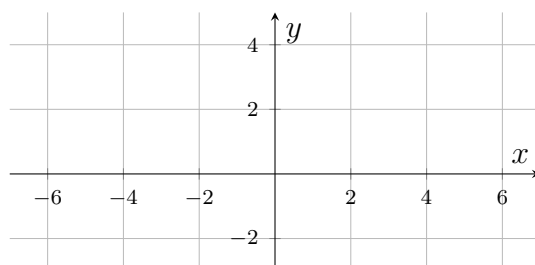
7. $G(0, 0)$ and $H(4, 3)$ are dilated about the origin, giving $G'(0, 0)$ and $H'(12, 9)$.

$GH =$ _____ $G'H' =$ _____ Scale factor = _____

8. $J(2, 1)$ and $K(6, 4)$ are dilated about the origin with scale factor 2, giving $J'(4, 2)$ and $K'(12, 8)$.

Slope of $JK =$ _____ Slope of $J'K' =$ _____

- 9.** A triangle with vertices $(2, 1)$, $(5, 1)$ and $(2, 3)$ is mapped onto the triangle with vertices $(-2, 1)$, $(-5, 1)$ and $(-2, 3)$. Use the grid if it helps.



First base = _____ Image base = _____

Name the transformation and explain why it makes the two triangles congruent.

- 10.** A right triangle has legs of 5 and 12 drawn from the origin to $(5, 12)$. A larger right triangle runs from the origin to $(15, 36)$.

Longest side, small = _____ Longest side, large = _____ Ratio = _____

11. Triangle ABC has $A(0,0)$, $B(4,0)$ and $C(0,3)$. It is dilated about the origin with scale factor 2, giving $A'(0,0)$, $B'(8,0)$ and $C'(0,6)$.

Slope of BC = _____ Slope of $B'C'$ = _____

Explain what the two slopes tell you about the angles of the two triangles.

12. Triangle ABC has $A(-4,-2)$, $B(0,-2)$ and $C(-4,1)$. Triangle DEF has $D(0,0)$, $E(6,0)$ and $F(0,4.5)$.

AB = _____ DE = _____ Ratio $DE : AB$ = _____

Are the two triangles congruent, similar but not congruent, or neither? Justify your choice using the numbers you found.
